Name of the Student..... Scholar Number.....

# MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
## DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

**EXAMINATION: Mid Term Examination** **MONTH and YEAR: October - November 2025**

**Course:** M.Tech **Semester:** I **Branch:** AC/CN/IS/AI **Subject Code:** CS AC 1166/CS CN 1154/CS IS 1154/CAI AI 1151
**Subject Name:** Web Search & Information Retrieval **Maximum Marks:** 20
**Duration:** 60 Minutes **Date:** 03/11/2025 **Time:** 09:30 AM To 10:30 AM

**Note:** *Answer all questions.*

| Q. No. | Questions | Marks | COs |
|---|---|---|---|
| 1 | **(A)** Explain the structure and purpose of an inverted index used in Boolean retrieval systems. How does it overcome the limitations of a term-document incidence matrix for large document collections? | 3 | CO1 |
| | **(B)** Consider the following document collection:<br>D1: "When Antony found Julius Caesar dead, he cried almost to roaring; and he wept when at Philippi he found Brutus slain."<br>D2: "Calpurnia! Peace, ho! Caesar shall forth. Cowards die many times before their deaths: The valiant never taste of death but once."<br>D3: "I did enact Julius Caesar: I was killed i' the Capitol; Brutus killed me."<br>D4: "Othello, the Moor of Venice, said: O, beware, my lord, of jealousy; it is the green-eyed monster which doth mock the meat it feeds on."<br>Consider the query given below. Describe step by step how this query is processed using postings lists in an inverted index (Build inverted index for terms in the query).<br>(a) Brutus AND Caesar AND NOT Calpurnia<br>(b) Caesar OR Calpurnia<br>(c) (Antony OR Brutus) AND NOT Othello | 2 | |
| 2 | **(A)** Why are stop-words like the, of, and often removed during indexing? Suppose your corpus has 100000 tokens and the following frequencies: Stop-words = 60000 tokens, Content words = 40000 tokens. If removing stop-words reduces index size by 70 %, estimate the new average posting-list length per term assuming the dictionary has 10000 terms. | 2 | CO2 |
| | **(B)** A postings list for the term **"machine"** contains the following 20 document IDs: [3, 7, 9, 11, 14, 17, 20, 25, 28, 31, 35, 40, 44, 49, 53, 60, 65, 70, 78, 81]<br>Another term **"learning"** has the postings list: [25, 49, 60, 81]<br>(a) If standard postings list intersection is used (no skips), compute the total number of comparisons needed to find the intersection.<br>(b) Now assume skip pointers are inserted every $\sqrt{P}$ positions (where P=20).<br>(i) Determine how many skip pointers will be created.<br>(ii) Using the skip pointers, estimate how many postings comparisons can be avoided, and compute the new total number of comparisons. | 3 | |
| 3 | **(A)** Explain analytically how a permuterm index enables efficient processing of both prefix (auto*) and suffix (*tion) queries. Also state one key limitation in terms of space or update cost. | 2 | CO2 |
| | **(B)** **(i)** Using the Levenshtein algorithm. Compute the edit distance between the words kitten and sitting. Backtrack and show what operations are performed.<br>**(ii)** If your retrieval system allows correction within edit distance $\leq 2$, determine whether kitten → sitting would be accepted. | 3 | |
| 4 | **(A)** For k = 3, list all 3-grams (use $ boundaries) for the words **"information"** and **"infomation"**. Compute Jaccard Coefficient. If the acceptance threshold is 0.7, determine whether these words will be accepted as a tolerant match. Explain how changing k from 3 to 4 affects Jaccard similarity and tolerance to spelling errors. | 2 | CO2 |
| | **(B)** **(i)** A search engine returns 20 documents, out of which 12 are relevant to the user's query. The total number of relevant documents in the collection is 30. Compute precision and recall for this query.<br>**(ii)** If the system is tuned to retrieve more documents (40 total, 18 relevant), compute the new precision and recall. Discuss the trade-off between precision and recall observed. | 3 | |

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# MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
## DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

**EXAMINATION: End Term Examination** **MONTH and YEAR: December 2025**

**Course:** M.Tech **Semester:** I **Branch:** AC/CN/IS/AI **Subject Code:** CS AC 1166/ CS CN 1154/CS IS 1154/ CAI AI 1162
**Subject Name:** Web Search & Information Retrieval **Maximum Marks:** 40
**Duration:** 2 Hrs **Date:** 10/12/2025 **Time:** 02:30 PM To 04:30 PM

**Note:** *Answer all questions.*

| Q. No. | Questions | Marks | COs |
|---|---|---|---|
| 1 | (A) Consider the Boolean query: {***network AND security***}. The postings for the two terms are:<br>network → [2, 5, 9, 14, 18, 21, 27, 33, 38, 42, 47, 53, 60, 66, 72, 79]<br>security → [5, 10, 18, 27, 33, 41, 53, 60, 73, 79]<br>(i) Assuming there are no skip pointers, list all document-ID comparisons performed when evaluating ***network AND security***, and compute the total number of comparisons.<br>(ii) Add skip pointers to the postings list of ***network***. Show clearly where each skip pointer goes. Then re-evaluate ***network AND security***, listing the comparisons in order and clearly indicating each time you follow a skip pointer. How many comparisons are saved compared to part (ii)?<br>(iii) Explain why adding too many skip pointers or placing them poorly can hurt rather than help performance. Use your example to justify your answer.<br>(iv) Suppose ***security*** becomes much longer than ***network*** in a larger collection. Explain which list you would prefer to add skip pointers to, and why, when evaluating network AND security. | 4 | CO2 |
| | (B) Write the Soundex algorithm and explain its role in phonetic-based information retrieval. Compute the Soundex codes for the terms "Katherine" and "Catherine", showing all intermediate steps. Conclude by evaluating whether Soundex correctly identifies these two spellings as phonetic variants and briefly discuss one case where Soundex may correctly match terms that do not sound alike. | 4 | |
| 2 | (A) Compute the Levenshtein edit distance between *planning* and *planetarium*. Backtrack through the table to obtain one optimal sequence of operations. Generate 2-gram bigram sets with **$** boundaries for both strings. If you must choose only one metric (edit distance or Jaccard) to filter spelling candidates for this pair, which one would you choose and why? | 4 | CO1 |
| | (B) Consider the following document collection:<br>Doc1: "GPS-based tractor systems for precision agriculture"<br>Doc2: "precision mapping using satellite imagery"<br>Doc3: "agriculture drones enhance imagery analysis"<br>Doc4: "GPS modules used for drone navigation systems"<br>Build the inverted index for the terms in the query and evaluate the following:<br>(i) (GPS AND precision) OR imagery<br>(ii) systems AND NOT agriculture<br>(iii) (drones OR satellite) AND NOT GPS<br>(iv) imagery OR (precision AND NOT drones) | 4 | |

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| 3 | (A) Consider the query fi*mo*er. What Boolean query on a bigram index would be generated for this query? Can you think of a term that matches the permuterm query, but does not satisfy this Boolean query?<br>(B) Discuss how the performance of an information retrieval system can be evaluated. State why is it important. There are two information retrieval systems, System A and System B and they are compared based on the same query which was applied to a collection of 5 million documents. System A retrieved 85,678 documents of which 9,825 are relevant and System B retrieves 62,450 documents, of which 4,210 are relevant to the given query. Compute the required performance measure for both System A and System B and state which one has higher performance | 4<br><br>4 | CO3 |
| 4 | Consider the following document collection:<br>Doc1: "smart phone camera lens phone"<br>Doc2: "laptop phone laptop screen smart"<br>Doc3: "camera lens zoom smart camera"<br>Doc4: "tablet screen phone tablet lens"<br>(i) Can tf-idf weights of a term in a document exceed 1?<br>(ii) Compute the top two scoring documents on the query "smart camera phone screen" using the **ntc.atc** weighting scheme. | 8 | CO4 |
| 5 | (A) Explain the intuition behind relevance feedback in information retrieval. Why does user-provided relevance information often improve retrieval effectiveness? Discuss how relevance feedback addresses vocabulary mismatch and query drift.<br>(B) Explain the concept of a crawler frontier. Describe how priority queues, heuristics, and URL scoring strategies influence the order in which pages are crawled, and discuss why naive FIFO crawling is ineffective on the modern Web. | 4<br><br>4 | CO4 |

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